

CA-N (NETWORK)

ASSIGNMENT 2

Identify a real world application for both parallel computing and networked system. Explain how these technologies are used and why they are important in that context.

A Real world application where both parallel computing and networked system are essential is cloud computing.

These technologies are used as -

1) Parallel Computing -

cloud computing platforms such as AWS, Google cloud, or Microsoft Azure, use parallel computing to break down large tasks into smaller sub-tasks and process them concurrently across multiple processors.

2) Networked system -

cloud computing platforms rely on networked system to connect distributed computing resources. These systems enable the communication between different servers or nodes that work together to complete a task.

Importance of These Technologies -

- **Scalability** - Parallel computing allows for the efficient processing of large datasets by dividing tasks and utilizing multiple cores. Networked systems enable this processing to happen across many machines, making it suitable... scalable.
- **Performance** - By combining parallelism and networking, cloud computing platforms can process tasks far faster than a single machine could.
- **Flexibility and Cost-efficiency** - Cloud platforms can offer pay as you go services which means businesses don't need to invest in large-scale hardware upfront. They can leverage networked systems to use as many computational resources as they need, scaling up or down depending on demand.